In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:
```
def f(x):
    return x**2 + 6*x+9
```

In [3]:
```
def df(x):
    return 2*x+6
```

In [4]:
```
def gradient_descent(starting_point, learning_rate, iterations):
    x = starting_point
    for i in range(iterations):
        x = x - learning_rate * df(x)  # update step
        print(f"Iteration {i+1}: x = {x:.4f}, f(x) = {f(x):.4f}")
    return x

starting_point = 2
learning_rate = 0.01
iterations = 100

minimum = gradient_descent(starting_point, learning_rate, iterations)
print(f"\nLocal minimum occurs at x = {minimum:.4f}, f(x) = {f(minimum):.4f}")
```

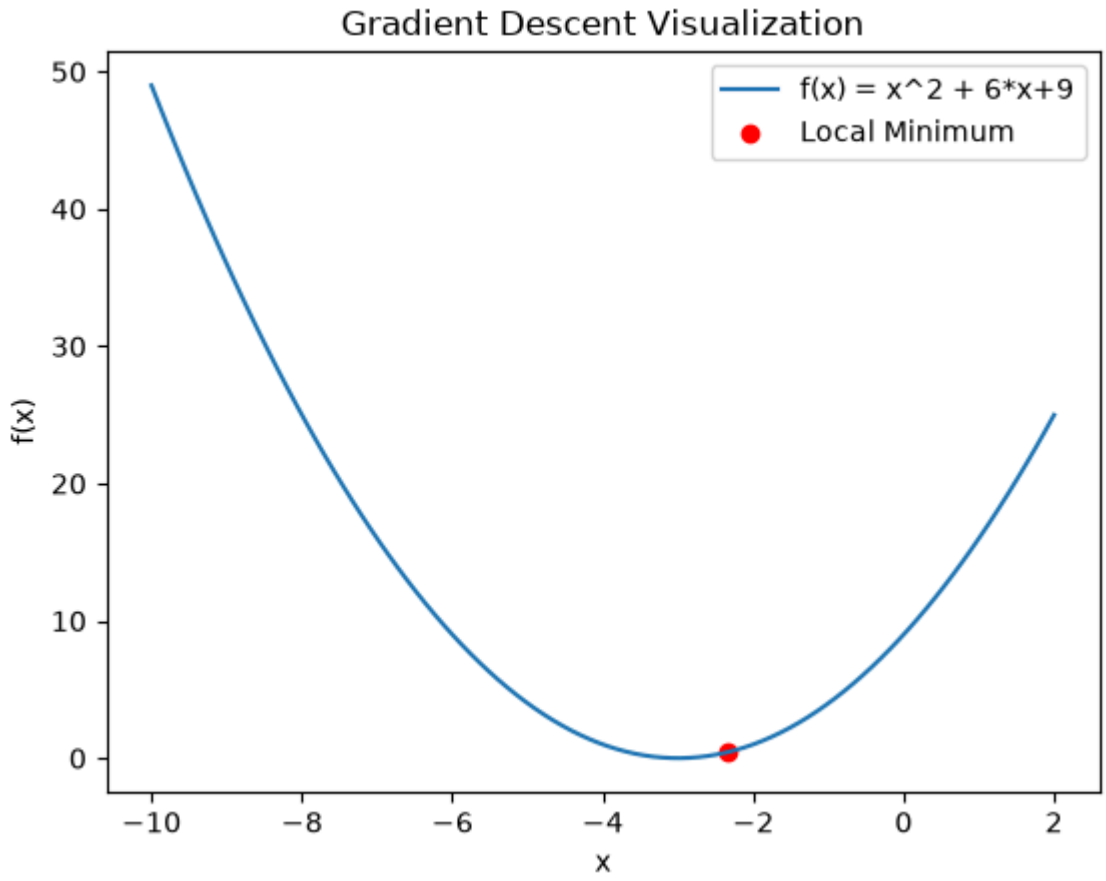
```
Iteration 1: x = 1.9000, f(x) = 24.0100
Iteration 2: x = 1.8020, f(x) = 23.0592
Iteration 3: x = 1.7060, f(x) = 22.1461
Iteration 4: x = 1.6118, f(x) = 21.2691
Iteration 5: x = 1.5196, f(x) = 20.4268
Iteration 6: x = 1.4292, f(x) = 19.6179
Iteration 7: x = 1.3406, f(x) = 18.8410
Iteration 8: x = 1.2538, f(x) = 18.0949
Iteration 9: x = 1.1687, f(x) = 17.3784
Iteration 10: x = 1.0854, f(x) = 16.6902
Iteration 11: x = 1.0037, f(x) = 16.0293
Iteration 12: x = 0.9236, f(x) = 15.3945
Iteration 13: x = 0.8451, f(x) = 14.7849
Iteration 14: x = 0.7682, f(x) = 14.1994
Iteration 15: x = 0.6928, f(x) = 13.6371
Iteration 16: x = 0.6190, f(x) = 13.0971
Iteration 17: x = 0.5466, f(x) = 12.5784
Iteration 18: x = 0.4757, f(x) = 12.0803
Iteration 19: x = 0.4062, f(x) = 11.6019
Iteration 20: x = 0.3380, f(x) = 11.1425
Iteration 21: x = 0.2713, f(x) = 10.7013
Iteration 22: x = 0.2059, f(x) = 10.2775
Iteration 23: x = 0.1417, f(x) = 9.8705
Iteration 24: x = 0.0789, f(x) = 9.4796
Iteration 25: x = 0.0173, f(x) = 9.1042
Iteration 26: x = -0.0430, f(x) = 8.7437
Iteration 27: x = -0.1022, f(x) = 8.3975
Iteration 28: x = -0.1601, f(x) = 8.0649
Iteration 29: x = -0.2169, f(x) = 7.7456
Iteration 30: x = -0.2726, f(x) = 7.4388
Iteration 31: x = -0.3271, f(x) = 7.1443
Iteration 32: x = -0.3806, f(x) = 6.8613
Iteration 33: x = -0.4330, f(x) = 6.5896
Iteration 34: x = -0.4843, f(x) = 6.3287
Iteration 35: x = -0.5346, f(x) = 6.0781
Iteration 36: x = -0.5839, f(x) = 5.8374
Iteration 37: x = -0.6323, f(x) = 5.6062
Iteration 38: x = -0.6796, f(x) = 5.3842
Iteration 39: x = -0.7260, f(x) = 5.1710
Iteration 40: x = -0.7715, f(x) = 4.9662
Iteration 41: x = -0.8161, f(x) = 4.7696
Iteration 42: x = -0.8597, f(x) = 4.5807
Iteration 43: x = -0.9026, f(x) = 4.3993
Iteration 44: x = -0.9445, f(x) = 4.2251
Iteration 45: x = -0.9856, f(x) = 4.0578
Iteration 46: x = -1.0259, f(x) = 3.8971
Iteration 47: x = -1.0654, f(x) = 3.7428
Iteration 48: x = -1.1041, f(x) = 3.5945
Iteration 49: x = -1.1420, f(x) = 3.4522
Iteration 50: x = -1.1792, f(x) = 3.3155
Iteration 51: x = -1.2156, f(x) = 3.1842
Iteration 52: x = -1.2513, f(x) = 3.0581
Iteration 53: x = -1.2862, f(x) = 2.9370
Iteration 54: x = -1.3205, f(x) = 2.8207
Iteration 55: x = -1.3541, f(x) = 2.7090
Iteration 56: x = -1.3870, f(x) = 2.6017
Iteration 57: x = -1.4193, f(x) = 2.4987
Iteration 58: x = -1.4509, f(x) = 2.3997
Iteration 59: x = -1.4819, f(x) = 2.3047
Iteration 60: x = -1.5122, f(x) = 2.2134
```

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Iteration 61: x = -1.5420, f(x) = 2.1258
Iteration 62: x = -1.5711, f(x) = 2.0416
Iteration 63: x = -1.5997, f(x) = 1.9608
Iteration 64: x = -1.6277, f(x) = 1.8831
Iteration 65: x = -1.6552, f(x) = 1.8085
Iteration 66: x = -1.6821, f(x) = 1.7369
Iteration 67: x = -1.7084, f(x) = 1.6681
Iteration 68: x = -1.7343, f(x) = 1.6021
Iteration 69: x = -1.7596, f(x) = 1.5386
Iteration 70: x = -1.7844, f(x) = 1.4777
Iteration 71: x = -1.8087, f(x) = 1.4192
Iteration 72: x = -1.8325, f(x) = 1.3630
Iteration 73: x = -1.8559, f(x) = 1.3090
Iteration 74: x = -1.8788, f(x) = 1.2572
Iteration 75: x = -1.9012, f(x) = 1.2074
Iteration 76: x = -1.9232, f(x) = 1.1596
Iteration 77: x = -1.9447, f(x) = 1.1137
Iteration 78: x = -1.9658, f(x) = 1.0696
Iteration 79: x = -1.9865, f(x) = 1.0272
Iteration 80: x = -2.0068, f(x) = 0.9865
Iteration 81: x = -2.0266, f(x) = 0.9475
Iteration 82: x = -2.0461, f(x) = 0.9099
Iteration 83: x = -2.0652, f(x) = 0.8739
Iteration 84: x = -2.0839, f(x) = 0.8393
Iteration 85: x = -2.1022, f(x) = 0.8061
Iteration 86: x = -2.1201, f(x) = 0.7741
Iteration 87: x = -2.1377, f(x) = 0.7435
Iteration 88: x = -2.1550, f(x) = 0.7141
Iteration 89: x = -2.1719, f(x) = 0.6858
Iteration 90: x = -2.1884, f(x) = 0.6586
Iteration 91: x = -2.2047, f(x) = 0.6325
Iteration 92: x = -2.2206, f(x) = 0.6075
Iteration 93: x = -2.2362, f(x) = 0.5834
Iteration 94: x = -2.2514, f(x) = 0.5603
Iteration 95: x = -2.2664, f(x) = 0.5381
Iteration 96: x = -2.2811, f(x) = 0.5168
Iteration 97: x = -2.2955, f(x) = 0.4964
Iteration 98: x = -2.3096, f(x) = 0.4767
Iteration 99: x = -2.3234, f(x) = 0.4578
Iteration 100: x = -2.3369, f(x) = 0.4397

Local minimum occurs at x = -2.3369, f(x) = 0.4397
```

In [5]:

```
x_vals = np.linspace(-10, 2, 100)
y_vals = f(x_vals)
plt.plot(x_vals, y_vals, label="f(x) = x^2 + 6*x+9")
plt.scatter(minimum, f(minimum), color='red', label="Local Minimum")
plt.xlabel("x")
plt.ylabel("f(x)")
plt.title("Gradient Descent Visualization")
plt.legend()
plt.show()
```


Gradient Descent Visualization
f(x) = x^2 + 6*x+9
Local Minimum
f(x)
x
50
40
30
20
10
0
−10
−8
−6
−4
−2
0
2

In [ ]: